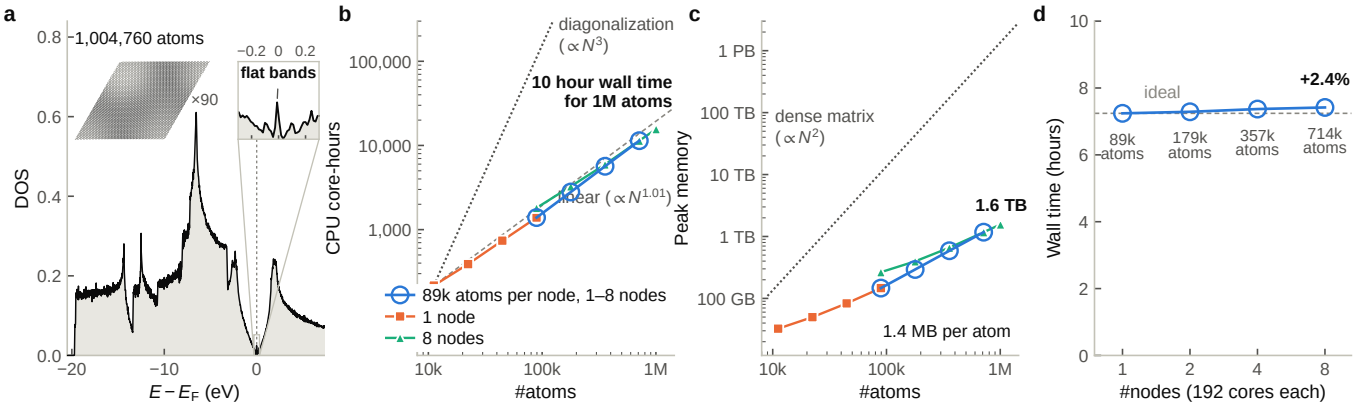

a
1,004,760 atoms
×90
−0.2 0 0.2
flat bands
DOS
0.8
0.6
0.4
0.2
0.0
−20
−10
0
E − E_F (eV)
b
diagonalization (∝N^3)
10 hour wall time for 1M atoms
linear (∝N^1.01)
CPU core-hours
100,000
10,000
1,000
89k atoms per node, 1–8 nodes
1 node
8 nodes
10k
100k
1M
#atoms
c
dense matrix (∝N^2)
Peak memory
1 PB
100 TB
10 TB
1 TB
100 GB
1.6 TB
1.4 MB per atom
10k
100k
1M
#atoms
d
ideal
+2.4%
Wall time (hours)
10
8
6
4
2
0
89k atoms
179k atoms
357k atoms
714k atoms
1
2
4
8
#nodes (192 cores each)